

Scientists working on chip to mimic human brain

Scientists from the Boise State University are working on a computer chip that has the ability to mimic the human brain. Despite its advancements and the massive speeds at which computer chips can function, none have come close to what the human brain is capable of. The team of scientists working on the project, including Kris Campbell, Indian-origin Vishal Saxena and Elisa Barney Smith, aims to develop a type of computer architecture that will work like a human brain.

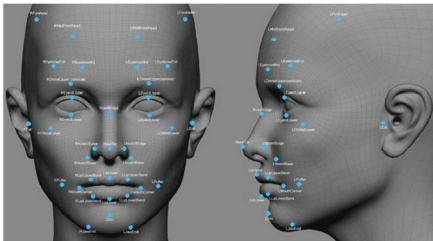
The main component of the project is memristor, a kind of resistor that has been programmed to different levels of resistance by the use of electrical pulses. This resistor can remember the value of the resistance even after the power is removed.

Researchers close to developing a 'thinking' chip

Researchers at the International Business Machines (IBM) are close to developing a silicon-based chip that would think and process like a mammal. The project, co-funded by the US military and many other academic organisations, is called Systems of Neuromorphic Adaptive Plastic Scalable Electronics (SyNAPSE).

The researchers have achieved a big success and finally been able to develop a complete neural networks development package, which is capable of processing and responding to any sensory input.

The team, including Dr Dharmendra S. Modha, a research fellow at IBM, and Rajit Manohar, VLSI expert at the Cornell University, has said that while the project does have military applications ranging from unmanned drones to facial recognition surveillance, they are more interested in the scientific and financial applications of this system.



3D facial recognition

3D facial recognition tech to catch criminals

Voice of Big Data (VOBD), which is a US-based data analytics and business intelligence solutions firm with offshore office in India, said it would launch 'Facial Signature', its flagship product, before the next quarter. VOBD asserts that the firm would be the first in India to launch a 3D facial detection product that could aid police and investigating officials to recognise criminals by using the data analytics tools.

"The product has already gained recognition in the US and Indian markets. Though we are yet to launch, we have three clients in the US. In India, we have helped a few police stations to solve their cases within a few hours," said VOBD chief executive Srinu Kishan. So far the company has invested \$200,000 in the project and is aiming at investing \$1,400,00 by March 2015.

The company stated that the Facial Signature is nearly 90 per cent accurate and the algorithms involved in this application can identify the 80 nodal points on a person's face. It will also be possible to identify a person in a disguise or wearing a 'burkha.'

A robot from China set to land on moon

Chang'e-3 is the name of China's endeavor to explore the rocky terrains on moon. The unmanned robot is all set to roll on moon's surface at the end of 2013. It marks the first of its kind attempt in the series of Chang'e robots designed to conquer the moon.

Learning from the past failed attempts, Chinese engineers are more cautious this time around and unique, slow-landing technique has been developed to place the bot on moon's surface safely. However, the details are yet to be revealed. The bot has successfully undergone the research and construction stage and is presently being tested for the launch. The Xichang Satellite Launch Center in Sichuan province has been nominated for the launch of the moon explorer.



China is preparing to launch Chang'e-3